



Indian Institute of Technology, Kanpur
Computer Science and Engineering

Quiz 4

CS670: Cryptographic Techniques for Privacy Preservation

Instructor: Adithya Vadapalli

06/11/2025

Name: _____

Roll Number: _____

This quiz contains 5 pages (including this cover page) and 7 questions. Total of marks is 25.

Distribution of Marks

Question	Points	Score
1	2	
2	6	
3	6	
4	2	
5	3	
6	3	
7	3	
Total:	25	

1. Describe briefly the differences between Information-Theoretic PIR and Computational PIR in terms of the following:

(a) (1 point) Number of Servers

Solution: IT PIR has multiple servers while C-PIR has only one servers.

(b) (1 point) Assumptions

Solution: IT-PIR: Non-Collusion among a subset of servers. C-PIR: Some Problem like LWE being a hard problem.

2. (6 points) In the FHE-based Computational PIR protocol, we discussed that a major part of the communication can be pushed to a preprocessing phase. How is this achieved? Is it a symmetric PIR protocol (and why)?

Solution: Since the matrix A is independent of the message, $D \cdot A$ can be computed before we know what we want to download.

3. (6 points) Consider an IT-PIR protocol with *eight* servers. Each of the eight servers holds a replica of the database. What is the minimum number of (2,2)-DPFs that need to be sampled so that even if only two servers respond, we can recover the message? How will you distribute the keys?

Solution: We need to sample 3 DPFs. We will use the binary representation of the index to distribute the DPF keys. Suppose that the 3 DPF Key pairs that are sampled are $(k_0^{(0)}, k_1^{(0)})$, $(k_0^{(1)}, k_1^{(1)})$, $(k_0^{(2)}, k_1^{(2)})$. An index whose binary representation is $b_1b_2b_3$ gets the keys $(k_{b_1}^{(0)}, k_{b_2}^{(1)}, k_{b_3}^{(2)})$

4. (2 points) Can we have a single-server information-theoretic PIR?

Solution: No

5. (3 points) Recall the Shamir Secret Sharing-based PIR protocol to download the j -th row. Suppose the database has n rows and s columns. What is the upload cost?

Solution: $O(s)$

6. (3 points) Recall the Shamir Secret Sharing-based PIR protocol to download the j -th row. Why is linearity of Shamir Secret Sharing a critical property?

Solution: Because of linearity, Lagrange interpolation after the dotproduct would yield the correct result.

7. (3 points) Recall the DPF-based PIR protocol. Is it a symmetric PIR protocol? Why?

Solution: Yes

This page is intentionally left blank to accommodate work that wouldn't fit elsewhere and/or scratch work.